Diksha Patkar

A Unified AI and Blockchain Framework

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A Unified AI and Blockchain Framework for Jasmine Disease Detection, Yield, Smart Irrigation, Harvest Planning, and Traceability

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Abstract - Jasmine is one of the economically important crops in South India where the income of farmers can be affected by diseases, poor irrigation and market uncertainty. This paper presents an integrated AI and blockchain framework for combining disease detection, yield prediction, smart irrigation and market aware harvest intelligence. The performance of three CNN architectures (EfficientNet-B0, ResNet50, and MobileNetV2) was evaluated on 7,325 images of jasmine leaves belonging to eight classes. The highest classification accuracy of 94.04% was achieved by EfficientNet-B0. SVM yield prediction resulted in R^2 of 0.8311. The Smart Irrigation Advisor uses classification to determine irrigation requirements and regression to estimate water quantities. The XGBoost model achieved an accuracy of 87.50% and a ROC-AUC of 94.95% for classifying irrigation needs, and an MAE of 0.89 L/plant, R2 of 0.8758, and MAPE of 4.46% for predicting irrigation quantities. The market-aware module also predicts the demand for jasmine, market price, buyer acceptance and unsold risk to support recommendations about harvest dates. Blockchain-based batch traceability and expert validation increase transparency and reliability further.

Index Terms - *Jasmine, plant disease classification, deep learning, EfficientNet, precision agriculture, yield prediction, Smart Irrigation, irrigation prediction, water management, blockchain traceability, machine learning.*

I. INTRODUCTION

Jasmine is an economically important flower crop in India, particularly in the southern states where it provides income for many farming households. However, diseases such as leaf blight, root rot and rust and insect pests can lead to reduced quality of flowers and crop yield. Disease detection is usually performed by manual inspection and experience of farmers, which is time-consuming and difficult to scale to large cultivation areas [11, 12]. Therefore, diseases may not be detected until visible symptoms are present, leading to significant crop losses.

Recent progress in deep learning has made possible automated plant disease classification through convolutional neural networks (CNNs) trained on leaf images [9, 10]. Other agricultural applications are crop-yield prediction, object detection for flower localization [17], image segmentation [16], and IoT-based crop monitoring [1-4]. However, these skills are often developed in isolation, limiting the capacity to provide comprehensive support for jasmine cultivation.

To bridge this gap, the proposed system integrates detection of jasmine disease, severity assessment, treatment advisory, yield prediction and expert validation. We compare three CNN architectures, EfficientNet-B0, ResNet50 and MobileNetV2 on an eight-class dataset of 7,325 images of jasmine leaves. The framework also estimates disease severity and suggests treatments. The yield-prediction module estimates the expected jasmine production based on agronomic and environmental parameters.

The proposed system includes a Smart Irrigation Advisor that calculates irrigation needs and quantifies water application based on soil, weather, crop-stage, irrigation-history and water-requirement parameters. It also includes a market-aware harvest intelligence module that predicts jasmine demand, market price, buyer-acceptable quantity, and unsold risk to recommend appropriate harvest dates based on expected market conditions. A hybrid on-chain/off-chain blockchain mechanism provides traceability of supply-chain at batch level with role based access control for different stakeholders. The system aims to provide a single platform for jasmine cultivation and distribution, including crop-health monitoring, yield prediction, irrigation decision support, market-aware harvest planning and traceability. Further independent field validation is necessary to evaluate its effectiveness in the field.

The main contributions of the work are as follows.

1. An integrated jasmine monitoring framework for disease classification, severity, treatment advisory, yield prediction and expert validation.
2. A comparative assessment of EfficientNet-B0, ResNet50 and MobileNetV2 for eight-class disease classification using 7,325 jasmine leaf images.

3. Random Forest, XGBoost and Support Vector Regression (SVR) based yield-prediction module on 16 agronomic input features.
4. A Smart Irrigation Advisor for classifying irrigation requirements and predicting irrigation quantities.
5. A market-aware harvest intelligence module that recommends harvest dates by predicting demand, price, buyer acceptance and unsold risk.
6. A hybrid blockchain based mechanism for batch level traceability and role based access control across jasmine supply chain.

II. LITERATURE REVIEW

A. Existing Machine Learning and Deep Learning Techniques

Recently many studies propose machine learning and deep learning frameworks for jasmine cultivation. LeafSpotNet [4] is a leaf spot disease detection model based on a modified MobileNetV3 classifier with the addition of Conditional Generative Adversarial Network (CGAN) augmentation and feature selection Particle Swarm Optimization (PSO). Another hybrid framework uses IoT sensor data (temperature, humidity, and soil moisture) and Random Forest for yield prediction and CNNs for health monitoring [5]. A custom-backbone UNet with DCGAN augmentation segments leaf-spot regions with a MobileNetV4 backbone [6] providing accurate disease localization. Other work looks at unified CNN and Random Forest pipelines for multi-level severity recognition [7]. FSCA-EUNet: A frequency-spatial cross attention model for fine-grained bloom-stage quality grading with small inter-class difference [8].

B. Review of previous works and their shortcomings

However, while these studies demonstrate good technical proficiency, there are a number of limitations that restrict their practical deployment. Most depend on curated or synthetic datasets, which may not be similar to real-world field conditions, such as extreme camera angles, shadows, or cluttered backgrounds [4, 6]. The non-linear characteristics of sensor data and computational expense of deep networks makes real-time deployment in the hybrid systems coupled with IoT challenging [5]. Most of the models are disease or growth stage specific, limiting their generalizability for holistic crop management [7, 6]. Also, complex architectures such as FSCA-EUNet are precise but need GPU resources not feasible on low-power mobile devices used in remote agricultural environments [8].

C. Research Gap and Our Answer

The literature highlights the gap that there are no common platforms for jasmine disease detection, severity estimation and farmer support. The proposed system tackles this by means of multi-class disease classification, severity classification in low, medium and high levels and CNN benchmarking. It offers treatment recommendations, yield prediction and market support via a FastAPI backend and Flutter mobile app. It also includes a Smart Irrigation Advisor, which uses soil, weather, crop and irrigation-history data to calculate irrigation needs and the water volume, and blockchain-based supply-chain traceability to help jasmine farmers.

Table 1. Comparison to Selected Prior Studies

Research	Methodology / Models	**Primary Focus**
LeafSpotNet (2024) [4]	MobileNetV3 + PSO + CGAN augmentation	Leaf Spot Classification (97% accuracy, F1 0.97)
Malathi et al. [5] (2024)	Random Forest(yield), CNN(health), Iot sensors	Predicting yields (89% accuracy) Monitoring health (F1 90.5%)
Custom Backbone UNet (2024) [6]	DCGAN, UNet + MobileNetV4	Disease region segmentation (0.91 PA, 0.95 mIoU)
Unified CNN-RF (2024) [7]	Random Forest + CNN features	Severity level recognition (91.79 % acc)
FSCA-EUNet (2025) [8]	Frequency-spatial attention U-Net like	Bloom stage quality grading (97.24% mAP, 95.52% precision)

III. METHODOLOGY

A. Suggested Approach

We present an AI based framework for detection, severity estimation, yield prediction and agricultural advisory of jasmine diseases. Jasmine leaf images are classified into healthy and diseased classes, they are split into training, validation and test sets and preprocessed by resizing, normalization, denoising, contrast enhancement and quality filtering. Data augmentation is done to increase the diversity of the training. In terms of accuracy, precision, recall, and F1-score, we evaluate three CNN models, EfficientNet-B0, ResNet50, and MobileNetV2 and select the best performing model for deployment. The predicted disease class helps in estimating the severity and recommending treatments. Agronomic and environmental parameters help in predicting the yield and Smart Irrigation helps to determine the irrigation requirements and estimate the quantity of water. These capabilities, along with market-aware harvest intelligence and blockchain-based supply-chain traceability, constitute the integrated framework. The overall pipeline is summarized as:

Data Collection → Preprocessing → Augmentation → Disease Classification → Severity Assessment → Yield Prediction and Smart Irrigation → Agricultural Advisory and Market-Aware Harvest Planning → Evaluation and Deployment.

B. Algorithms and Models

1) Data Collection and Preprocessing

Images of jasmine plants under diverse environmental and disease conditions were collected and annotated according to disease category, and divided into training, validation and test subsets. Images were preprocessed using resizing to fit the input size, color normalization to reduce variation due to devices and lighting, denoising and contrast enhancement to enhance features associated with diseases, and removal of low quality, blurred or duplicate images.

2) Data Augmentation

To increase the diversity of the training set and reduce overfitting, we applied standard geometric transformations (rotation, flipping, cropping, and scaling) and photometric transformations (brightness, contrast, and hue). Also, generative approaches, in particular CGAN[18], were considered for class-specific sample synthesis.

3) Feature Extraction

EfficientNet-B0 [15] was selected as the main backbone due to its compound scaling property, which achieves a trade-off between accuracy and computational efficiency, and is suitable for deployment on resource-constrained devices. ResNet50 [13] was chosen as a benchmark owing to its residual connections that facilitate the training of deep networks. MobileNetV2 [14] was chosen as a lightweight alternative that is optimized for faster inference and lower computational cost on mobile devices.

4) Disease Classification

Each backbone extracts hierarchical visual features from the preprocessed images and predicts the disease category it belongs to from the predefined disease categories. The performance is compared using accuracy, precision, recall and F1 score to select a most suitable model, the output of which is input to severity estimation and advisory modules.

5) Disease Severity Estimation

Besides binary detection, the system quantifies the infection extent by analyzing the segmented disease-affected region relative to the total leaf or plant area. It employs an encoder-decoder segmentation approach in the U-Net [16] style to segment the boundaries of the lesions. Each case is ranked as low, medium or high severity based on estimated area of impact and model confidence, which informs treatment recommendations.

6) Yield Prediction

A special regression module is used to estimate the effect of disease and environmental conditions on jasmine production, with the type of disease, severity, crop information and environmental parameters serving as inputs.

7) Advisory and Recommendation System

The advisory module gives disease-specific treatment suggestions, preventive measures and chemical/organic alternatives, based on the predicted disease, severity, crop information and environmental context. The AI-generated report can then be sent to an agronomist for validation before delivery to the farmer, in the expert-assisted workflow

8) Smart Irrigation Advisory

The system has a Smart Irrigation Advisor that calculates irrigation needs and provides an estimated recommended volume of water in litres per plant. It uses soil, weather, crop-stage, irrigation-history, and water-requirement variables such as soil moisture, rainfall, temperature, crop coefficient (Kc), previous irrigation, evapotranspiration (ET_0 and ETc), and water deficit. The irrigation requirement is modelled as a binary classification problem using Random Forest, XGBoost, Logistic Regression and SVM while the irrigation quantity is predicted using regression with Random Forest, XGBoost, SVR and Gradient Boosting Regression. The models are evaluated using metrics for performance and the best performing model is selected and deployed for each task.

9) Harvest-Time and Market Demand Prediction

The proposed system has a market-aware module that integrates to predict demand for jasmine, market price, quantity acceptable to buyers, risk of unsold and appropriate harvest date. It takes into account historical market trends, the intensity of festivals, previous yield and quality of flowers, weather, buyer capacity, transport distance, crop maturity, the severity of diseases and calendar features. Demand and price prediction was done using Random Forest, XGBoost and SVR with MAE, RMSE, R2 and MAPE. Un-sold risk was classified as Low, Medium or High. The dataset was divided into 80% for training and 20% for testing, and the corresponding predictions were aggregated to suggest an appropriate harvest date.

10) Model Evaluation

The models are evaluated according to accuracy, precision, recall and F1-score. Confusion matrix is used to analyze the class-wise errors. Inference time is also noted to check its suitability for real-time, on-device applications in agriculture. The final model is selected based on a blend of factors that include predictive performance, computational efficiency, and suitability for deployment.

IV. SYSTEM STRUCTURE

The farm and leaf-image data collected through the mobile application are preprocessed and analysed by an EfficientNet-based feature extractor and softmax classification layer. Farm and environment data are used for yield prediction. The advisory module integrates disease predictions, relevant weather and GIS data to give treatment recommendations. The generated report can be submitted for expert validation. The framework also includes smart irrigation, market aware harvest intelligence and blockchain based supply-chain traceability for holistic decision support.

A. Data set

A.1 Disease Detection Dataset

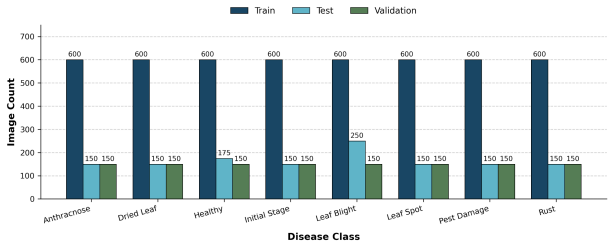


Figure 1. *Class-wise distribution of the jasmine leaf disease dataset into training, validation and testing datasets.*

The disease-classification dataset contains 7,325 images of jasmine leaves of eight classes: Healthy, Anthracnose, Dried Leaf, Initial Stage Disease, Leaf Blight, Leaf Spot, Pest Damage, and Rust Disease. Images were gathered from publicly available sources and manually labelled and split into 4,800 training, 1,200 validation and 1,325 testing images. Preprocessing was done by resizing, normalisation,

denoising and enhancement, and data augmentation was done by rotation, flipping and scaling to increase the diversity of images and reduce overfitting.

A.2 Smart Irrigation and Yield Prediction Dataset

An agricultural dataset with 20,000 records was taken for yield prediction and Smart Irrigation. It contains crop, soil, weather, irrigation, fertiliser, and plant-health conditions. Three targets were modelled separately, jasmine yield (kg/acre), irrigation requirement (Yes/No) and recommended irrigation quantity (litres per plant). Separate machine learning algorithms were trained for each task using relevant input features and tested using proper classification and regression metrics.

A.3 Demand Prediction Dataset

The Jasmine Harvest-Time and Market Demand Prediction module was tested against a synthesized dataset of 10,000 records with agricultural, weather and quality and market related features. The dependent variables were demand, market price, quantity accepted by buyers, unsold quantity and unsold-risk category. The dataset was partitioned into 80% training (8,000 records) and 20% testing (2,000 records) and the models were assessed using relevant regression and classification metrics. Since the dataset is synthetic, validation with real-world jasmine market data is needed to evaluate practical effectiveness.

B. Details of implementation

The system comprises three coordinated workflows: the Farmer, Admin and Expert (agronomist) workflows, each accessed through a dedicated interface.

Farmers application. After onboarding (choosing a language - Hindi, Marathi, Tamil, Kannada or Malayalam), farmers register via mobile-number authentication and land on a dashboard with farm details, disease-risk status and quick actions like "Scan Crop Now." The trained EfficientNet-B0 model takes in the captured or uploaded leaf images, processes and analyzes them, and displays the predicted disease class, confidence score. Farmers can also access treatment regimens that include chemical and organic options, dosage, timing, cost and safety precautions. They can also access disease information, a pest and disease library organized by growth stage, a dosage calculator and a marketplace for agricultural inputs.

Admin panel. The web dashboard provides a summary of platform activity, images uploaded, pending reviews, diseases detected and high severity alerts. The administrator can browse uploaded images and AI analysis results, manage inventory, moderate web-facing content, and generate consolidated disease-distribution reports. Unverified reports may be sent to agronomists.

Expert module. Agronomists monitor the farmers assigned to them, review the uploaded image and the AI-generated diagnosis ("Observe") and generate a validated report with the confirmed diagnosis, a custom treatment plan, and preventive measures ("Report") which is sent back to the farmer, completing the human-in-the-loop validation process.



Figure 2. *Flow of the system implementation*

V. IMPLEMENTATION

A. Technologies Used

The system consists of a farmer-facing mobile app built using Flutter/Dart and a responsive admin dashboard built using React/JavaScript. The backend is implemented in Python and exposes REST APIs (mostly via FastAPI) to connect the application, dashboard and AI models. The CNN models are trained using TensorFlow/Pytorch and the farmer profiles, farm records, scan history and inventory data are stored in MongoDB. Images are stored and served from Cloudinary. External integrations include OpenWeather for weather alerts, a GIS API for farm data based on location, and an API for support of regional languages.

1) Blockchain-Based Traceability for Jasmine Produce

The proposed system uses blockchain and MongoDB to provide transparent and tamper-proof traceability across the supply chain of jasmine. Instead of monitoring flowers one by one, each harvest batch gets a unique Batch ID (e.g. JAS-2026-001), connecting its cultivation data, AI disease findings, treatment records, harvest attributes, and timestamped supply chain events. It uses a hybrid on-chain/off-chain architecture where MongoDB stores the detailed records and images and the traceability events and cryptographic references (e.g. SHA-256 hashes) are stored in the blockchain to provide data integrity and verification.

Role-based access control (RBAC).
Table 2 summarizes the permissions granted to each stakeholder class on the blockchain ledger.

Table 2. Role-Based Access Control on the Traceability Ledger

Stakeholder	Permitted Actions
Farmer	Record cultivation details, disease-detection results, applied treatment, and harvest data
Expert/Agronomist	Validate/annotate disease diagnosis and treatment recommendations
Distributor	Log collection and transportation events
Seller	Record sale transactions and final distribution points
Customer	Read-only access to complete batch traceability history

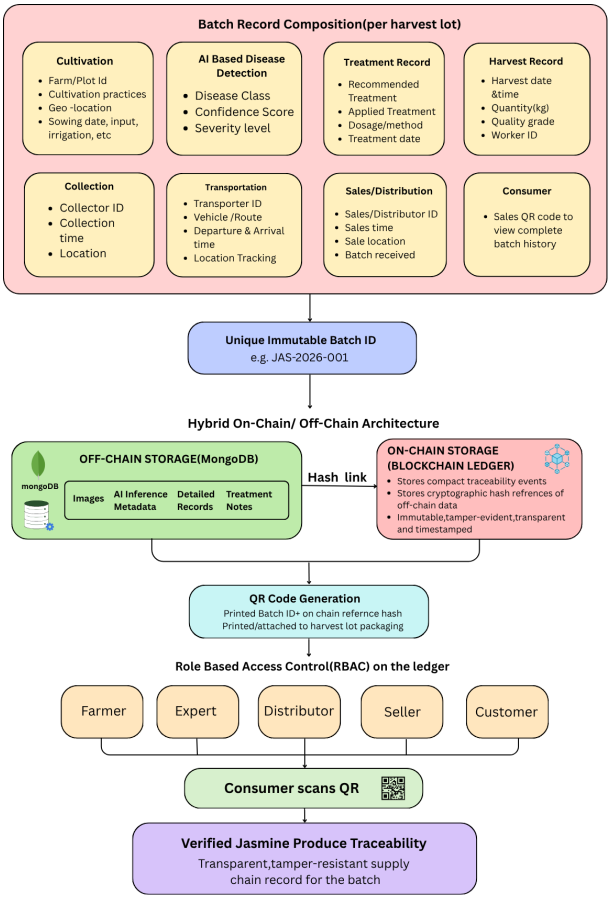


Figure 3. *Blockchain QR Code-based traceability flow*

2) Smart Irrigation Module

The Smart Irrigation Advisor uses parameters such as soil, weather, crop-stage, irrigation-history and water-requirement to determine irrigation needs and provides an estimate of water quantity (litres per plant) For classification, Random Forest, XGBoost, Logistic Regression and SVM were evaluated. For quantity prediction, Random Forest, XGBoost, SVR and Gradient Boosting Regression were tested. These selected models are used in the backend to give irrigation decisions and water recommendations to the farmer-facing application.

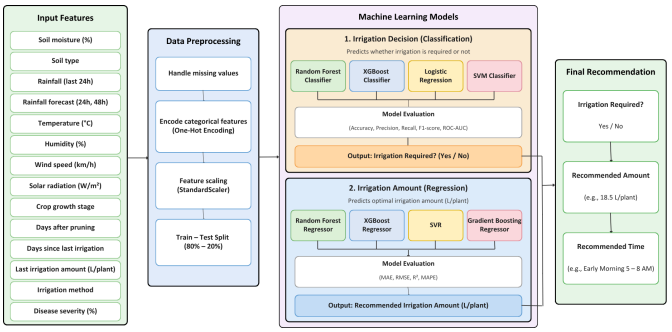


Figure 4: *Smart Irrigation Module*

3) Yield Prediction Module

The system has a yield-prediction module that predicts jasmine production (kg/acre) based on 16 agricultural features pertaining to crop, soil, weather, irrigation, fertiliser, plant health, and historical yield. The module is trained on the dataset of 20,000 records.

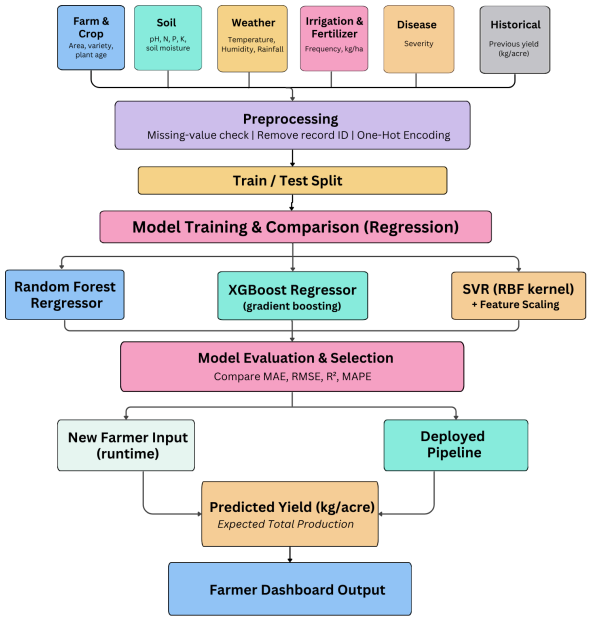


Figure 4. *Yield Prediction flow*

4) Market-Aware Harvest Intelligence Module

The system also has a market-aware harvest intelligence module to predict jasmine demand, price, buyer acceptance and unsold-flower risk based on crop, weather, seasonal and market features. Numerical features are standardised and categorical inputs are one-hot encoded. The demand and price prediction is validated using Random Forest [19], XGBoost [20] and RBF-kernel SVR [21] with 80:20 train-test split. Random Forest regression is used to predict buyer acceptance, and Random Forest classification is used to classify unsold risk into low, medium and high. These predictions are combined into a Harvest Suitability Score

$$HS = 0.35D + 0.30P + 0.25A + 0.10R$$

where D, P, A, and R represent the normalised demand, price, buyer acceptance, and low-risk score respectively. The highest-scoring date is recommended for harvest, with quantity based on predicted demand and buyer capacity.

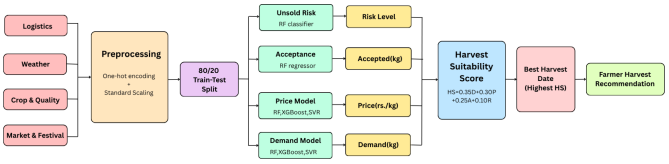


Figure 5: *Market-Aware Harvest Intelligence Module*

VI. RESULTS AND DISCUSSION

A. Disease Classification Performance

The proposed classification pipeline was evaluated on the jasmine leaf-image dataset (7325 images across eight classes; 4800 training, 1200 validation, and 1325 test images). Before training, images were resized to a consistent resolution and pixel-normalized; rotation, horizontal flipping, and zooming were applied as augmentation, and images were organized into class-wise directories. We trained and evaluated three architectures (EfficientNet-B0, ResNet50 and MobileNetV2.

Table 3. Classification Performance of the Deep Learning Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
MobileNetV2	92.08	94.20	92.08	91.85
ResNet50	92.60	94.18	92.60	92.32
EfficientNet-B0	94.04	95.00	96.00	94.00

The confusion matrix for EfficientNet-B0 on 1,325 test images (Fig. 7) demonstrates that most predictions are on the main diagonal, consistent with the test accuracy of 94.04%.

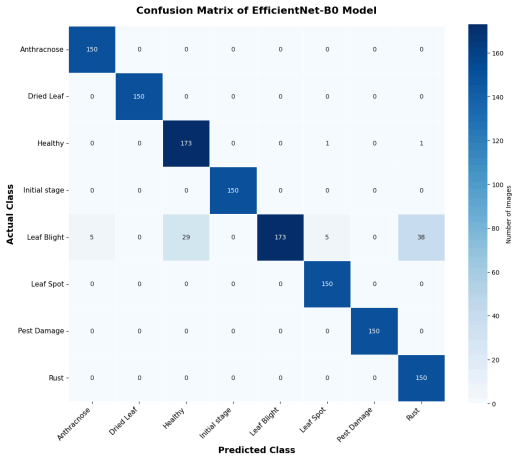


Figure 7. *Confusion matrix for the EfficientNet-B0 model for jasmine leaf disease classification on the held-out test set.*

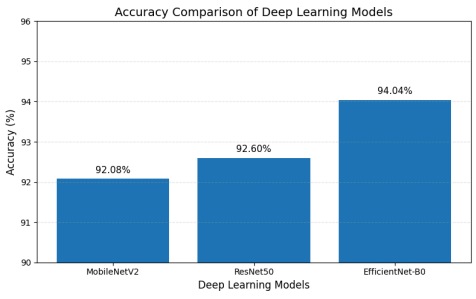


Figure 8. *Accuracy comparison of MobileNetV2, ResNet50, and EfficientNet-B0.*

Among the three architectures, the EfficientNet-B0 obtained the highest accuracy and was therefore selected as the main model for the proposed system. ResNet50 and MobileNetV2 had similar, slightly lower accuracy.

B. Yield Prediction Performance

The proposed framework also includes a module for predicting jasmine yield based on agricultural parameters. Regression was performed using ML algorithms such as Random Forest, XGBoost, and SVR (with a radial basis function (RBF) kernel). The performance of different algorithms was compared using MAE, RMSE, R^2 and MAPE. The results of the comparison are shown in Table 4 below.

Table 4. Comparison of Regression Models for Yield Prediction

Model	MAE (kg/acre)	RMSE (kg/acre)	R^2	MAPE (%)
Random Forest	84.79	104.99	0.7472	8.37
XGBoost	68.94	86.63	0.8279	6.74
SVR (RBF)	67.94	85.81	0.8311	6.63

Given the results shown in Fig. 9, among the yield-prediction models evaluated, SVR had the lowest MAE and RMSE and the highest R^2, thus was the best-performing model.

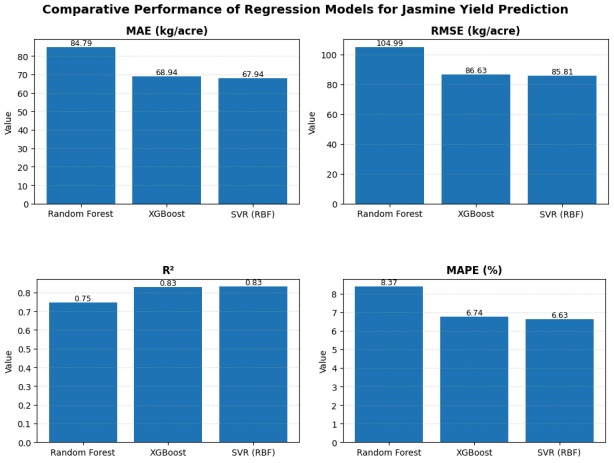


Figure 9. *Error Comparison of Random Forest, XGBoost and SVR*

C. Smart Irrigation Performance

The Smart Irrigation Advisor was evaluated as a classification task for determining whether irrigation was required. Four machine-learning algorithms, namely Random Forest, XGBoost, Logistic Regression, and SVM, were compared using accuracy, precision, recall, F1-score, and ROC-AUC.

Table 5: Comparative Performance of Classification Models for Irrigation Requirement Prediction

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	ROC-AUC (%)
Random Forest	87.00	87.96	84.48	86.18	94.46
XGBoost	87.50	87.85	85.83	86.83	94.95
Logistic Regression	86.20	85.33	86.04	85.68	94.43
SVM	83.45	82.59	83.02	82.81	91.75

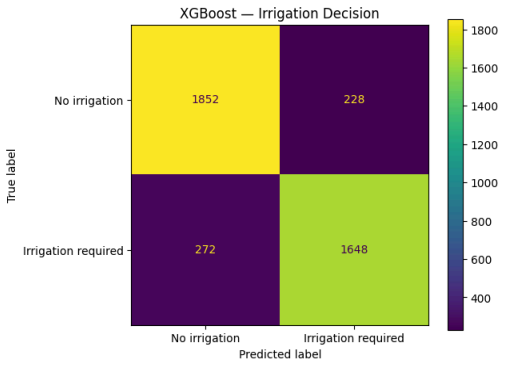


Figure 10: *Confusion Matrix of XGBoost for Irrigation-Requirement Classification.*

Table 6: Comparative Performance of Regression Models for Irrigation Quantity Prediction

Model	MAE (L/plant)	RMSE (L/plant)	R^2	MAPE
Random Forest	0.9982	1.2854	0.8409	5.0652
XGBoost	0.8854	1.1358	0.8758	4.4610
SVR	1.1098	1.4004	0.8112	5.5262
Gradient Boosting	0.9037	1.1397	0.8749	4.5150

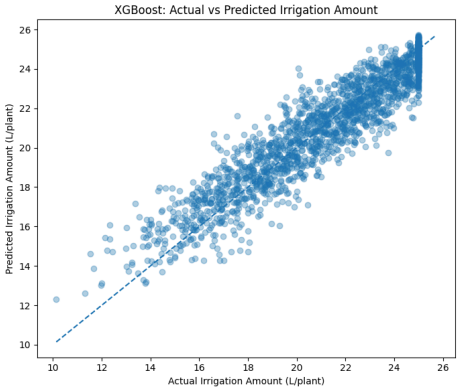


Figure 11: *Actual vs. predicted irrigation quantity using the best regression model*

As shown in Table 6, the best overall performance for the irrigation-quantity prediction was achieved by XGBoost with the MAE, RMSE, R^2 and MAPE of 0.8854 L/plant, 1.1358 L/plant, 0.8758 and 4.4610%, respectively.

D. Traceability based on blockchain

The implemented blockchain traceability module assigns a unique Batch ID to each jasmine harvest batch and records important events throughout the product lifecycle. For the demonstrated batch, seven sequential events were recorded: cultivation, AI-based disease detection, treatment, harvesting, collection, transportation, and sale. Each event contains relevant metadata along with a timestamp, data hash, previous hash, and block hash.

```
Adding 7 events to batch 'JAS-2026-001'...

  Added: CULTIVATION        by Ramesh Kumar           (block #0, hash=6aa7ddc41d55...)
  Added: DISEASE_DETECTION  by AI Diagnosis System    (block #1, hash=56239102f50e...)
  Added: TREATMENT          by Dr. Anjali Sharma      (block #2, hash=7588c1ec4035...)
  Added: HARVEST            by Ramesh Kumar           (block #3, hash=11864c5cb0f2...)
  Added: COLLECTION         by Suresh Logistics Pvt Ltd (block #4, hash=7954d1fad548...)
  Added: TRANSPORTATION     by Suresh Logistics Pvt Ltd (block #5, hash=53dff3a3b132...)
  Added: SALE               by Green Leaf Retail Store  (block #6, hash=0c85cd4d557d...)
```

Figure 12. *Successful recording of seven sequential traceability events for a jasmine harvest batch.*

The blockchain-based traceability system consists of seven blocks for the main events of the agricultural supply chain: Block 0 records the farming done by farmers. Block 1 records the disease detection done by AI. Block 2 records the treatment done by agricultural experts. Block 3 records the harvesting done by farmers. Block 4 records the collection. Block 5 records the transportation done by the logistics provider. Block 6 records the sale done by the retailer. Each block contains its hash value and the hash value of the preceding block. This feature helps to detect tampering and guarantees data integrity.


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"batch_id": "JAS-2026-002",
"events": [
  {
    "index": 0,
    "batchId": "JAS-2026-002",
    "eventType": "CULTIVATION",
    "role": "farmer",
    "actorId": "farmer_102",
    "actorName": "Ramesh Kumar",
    "timestamp": "1788688527.5053992",
    "offChainRef": "6a9d388f526ce23d893e185a",
    "dataHash": "0ba4581749c15ce7c3dc68e381cde3c555fd098a853334b96df44a8eac0ef943",
    "previousHash": "0000000000000000000000000000000000000000000000000000000000000000",
    "blockHash": "515e8445ed636a9adeb1a9826bd2297c59f6c1918e0227610433e39d97dfdff3"
  },
```


Figure 13. *Blockchain event record containing batchID, event metadata, timestamp, data hash, previous hash, and block hash.*

E. Harvest-Time and Market Demand Prediction Results

Table 9. Demand Prediction Results

Model	MAE	RMSE	R^2	MAPE (%)
XGBoost	12.734	16.006	0.850	11.124
Random Forest	12.731	16.089	0.849	11.170
SVR	13.863	17.407	0.823	12.074

XGBoost achieved the best demand-prediction performance with R^2 of 0.850 and MAE of 12.734. Random Forest performed comparably and SVR performed less well.

Table 10. Price Prediction Results

Model	MAE	RMSE	R^2	MAPE (%)
XGBoost	28.5400	31.6946	0.3154	19.0967
Random Forest	28.7847	31.8838	0.3072	19.3368
SVR	29.4742	29.4742	0.1648	19.3293

XGBoost had the best performance in price prediction with a R^2 of 0.3154 and MAE of 28.5400. Random Forest gave similar results and SVR had the lowest R^2 (0.1648).

Buyer Acceptance: XGBoost performed best in buyer acceptance prediction, achieving an MAE of 20.5191, MAPE of 35.2417%, and an R^2 score of 0.0231, indicating limited predictive capability and the need for additional features and real-world market data. For unsold-risk classification, Random Forest achieved the highest accuracy of 55.05%, while XGBoost obtained the highest F1-score of 0.5290. These results indicate scope for further model optimization and validation to improve prediction reliability.

Table11. Optimal Harvest-Time Recommendation

Day	Demand (kg)	Price (Rs/kg)	Acceptance (kg)	Risk	Suitability
1	107.42	166.38	81.15	Medium	0.628
2	185.88	166.38	81.15	Medium	0.925
3	134.38	166.38	81.15	Medium	0.730
4	119.83	166.38	81.15	Medium	0.675
5	80.36	166.38	81.15	Medium	0.525
6	113.76	166.38	81.15	Medium	0.652
7	109.61	166.38	81.15	Medium	0.636

The highest harvest suitability score was on Day 2 (0.925) followed by Day 3 (0.730) and the lowest score was on Day 5 (0.525). The recommendation is a blend of demand, price, buyer acceptance and unsold risk.

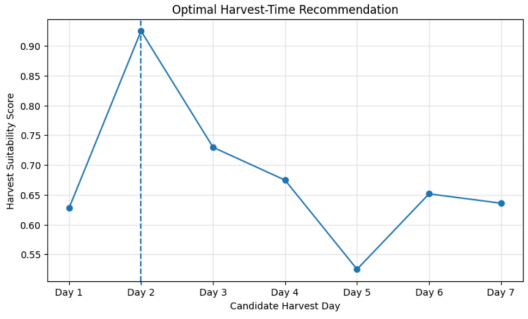


Figure 14. *AI-Based Harvest-Time Optimization and Decision Support.*

VII. CONCLUSION

In this paper, we propose a novel integrated AI and blockchain-enabled framework for jasmine disease detection, yield prediction, smart irrigation, market-aware harvest planning, and supply-chain traceability. The highest test accuracy of 94.04% was achieved by EfficientNet-B0 for eight-class leaf-condition classification and the best yield-prediction performance ($R^2 = 0.8311$) was obtained by SVR. For smart irrigation, XGBoost attained 87.5% classification accuracy and 0.8758 of R2 for prediction of irrigation-quantity. Tested on 10,000 synthetic records, market-aware harvest intelligence showed R^2=0.85 in demand prediction, and Random Forest achieved 55.05% accuracy in unsold-risk classification. It was found that day 2 was the best candidate for harvest with a score of 92.5%. Overall, our system shows promise for integrating AI-based agricultural decision support with blockchain-based traceability in a single platform. However, the reliability and generalisability of the proposed framework need to be established by validation using real-world field and market data.

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